Assignment 2: Git and Stream Temperature - August Analysis

Grace Bryan

2026-05-07

Data Loading

```
library(tidyverse)
library(lubridate)

# Load saved data

temp_data <- read.csv("data/temp_data.csv")

# Select and rename relevant columns
temp_data <- temp_data %>%
  select(
    agency_cd = monitoring_location_id,
    site_no = monitoring_location_id,
    Date = time,
    temp_C = value )

# Clean site names
temp_data$site_no <- gsub("USGS-", "", temp_data$site_no)

# Ensure proper format
temp_data$Date <- as.Date(temp_data$Date)
temp_data$month <- month(temp_data$Date, label = TRUE)
temp_data$year <- year(temp_data$Date)
```

Filter for August

```
aug_data <- temp_data %>%
  filter(month == "Aug")
```

Model: Temperature Trend Over Time

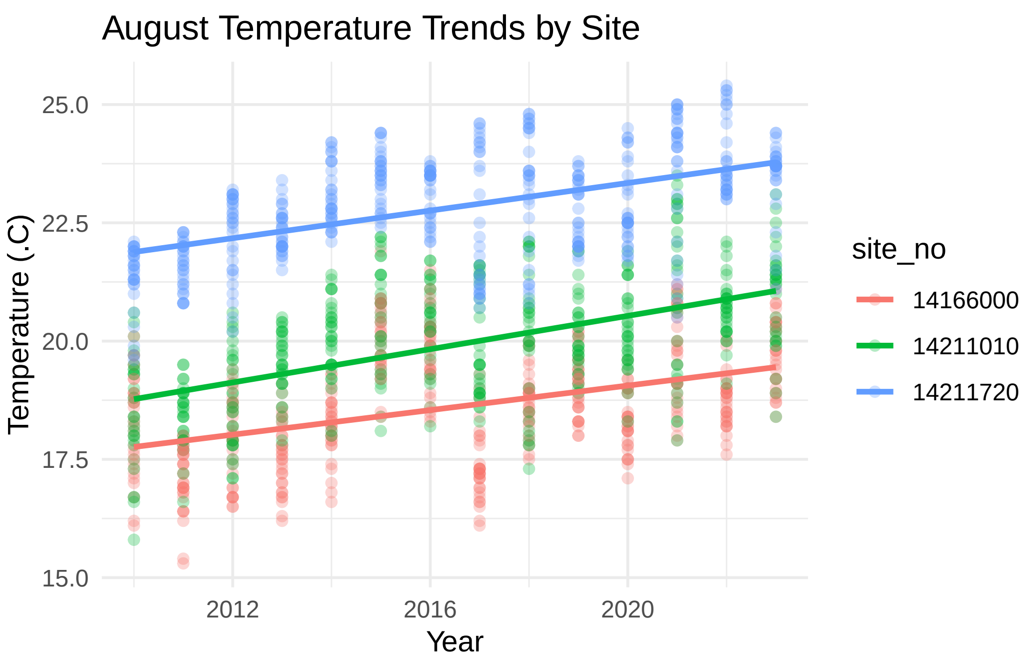
We use a generalized liner model (GLM) to estimate whether stream temperature is changing over time and whether trends differ across sites.

```
aug_model <- glm(temp_C ~ year * site_no,
                 data = aug_data)

summary(aug_model)
```

```

Call:
glm(formula = temp_C ~ year * site_no, data = aug_data)

Coefficients:
                       Estimate Std. Error t value Pr(>|t|)    
(Intercept)          -243.03103   25.50634  -9.528  < 2e-16 ***
year                    0.12975    0.01265  10.258  < 2e-16 ***
site_no14211010       -91.98714   36.01709  -2.554  0.01076 *  
site_no14211720       -28.41636   36.08229  -0.788  0.43111    
year:site_no14211010    0.04627    0.01786   2.590  0.00969 ** 
year:site_no14211720    0.01619    0.01789   0.905  0.36581    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for gaussian family taken to be 1.121551)

    Null deviance: 5985.3  on 1298  degrees of freedom
Residual deviance: 1450.2  on 1293  degrees of freedom
AIC: 3843.4

Number of Fisher Scoring iterations: 2
```

###*Visualization of Trends*

```r
ggplot(aug_data, aes(x = year, y = temp_C,
                      color = site_no)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    title = "August Temperature Trends by Site",
    x = "Year",
    y = "Temperature (˚C)" ) +
  theme_minimal()
```



Interpretation of Results

August stream temperatures show a clear increasing trend over time. The coefficient for year is positive and highly significant ($p < 0.001$), indicating that temperatures are rising at a rate of approximately 0.13°C per year at the baseline site (site 14166000).

There are also differences in warming rates between sites. Site 14211010 shows a significantly greater increase in temperature over time compared to the baseline site, with an additional warming rate of about 0.046°C per year. This suggests that this site is warming faster than the baseline.

In contrast, site 14211720 does not show a statistically significant difference in warming rate compared to the baseline site. Therefore, there is no strong evidence that this site is warming at a different rate.

Overall, the results indicate that August stream temperatures are increasing over time, with some variation in warming rates across sites.